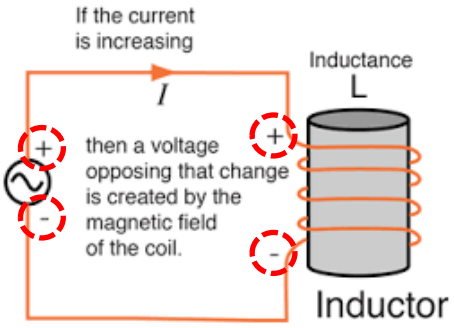
Chapter 49 - Electromagnetic Induction in Circuits
Calculus Physics II

Induction – Given a coil of wire, an emf can be created by a changing magnetic flux through the inside of the coil either from an external source or self-generated.

Inductance – The opposition to a changing current in a coil of wire due to an induced emf in that coil. The inductance of a coil of wire depends on the current, magnetic flux and coils creating the magnetic field at the center of the coil in which the emf is being induced.

- Inductance is measured in Henry's (H) 1 H = 1 Vs/A.
- The inductance acts to create an emf that opposes the applied external emf.
- A higher inductance gives a higher induced emf.

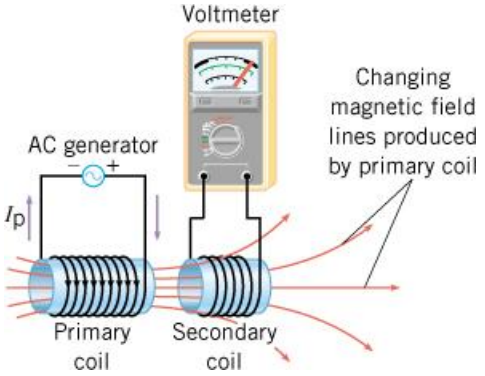


The diagram shows an ac power supply in series with an inductor. At the instant in time shown, when the power supply is positive the current will flow clockwise in the circuit as shown. The induced voltage across the inductor opposite this flow and sets up a induced voltage that would try to move the current counterclockwise with the overall effect of limiting the current flow in the circuit.

Mutual Inductance – The process by which a changing current in a coil of wire produces a changing magnetic field, when a second coil of wire is placed in this changing magnetic field an induced potential will be induced in the second coil due to the changing magnetic field (changing Flux) produced by the first coil. The induced potential will create an induced current in the secondary if there is a closed circuit.

Primary coils – The wire coils that produce the changing magnetic field by changing current is "seen" by the secondary coil. This coil is the source of energy that is transferred to the secondary coil.

Secondary coil – The wire coils that the emf is induced due to the changing magnetic field produced by the Primary coil and the energy from the primary coil is transferred to the secondary coil.

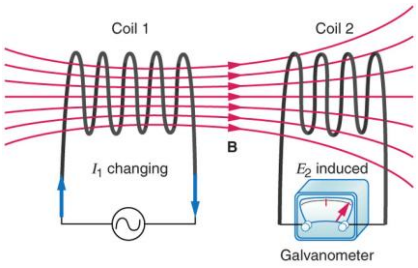


The mutual inductance is given by the symbol M and has units of Henery's (H). The mutual inductance depends on the number of loops in the secondary, the magnetic flux though the secondary's coils and the current in the primary.

$$V_{emf}\ (\text{Secondary}) = -N_s\left[\frac{d\Phi}{dt}\right] = -M\left[\frac{dI_p}{dt}\right]$$

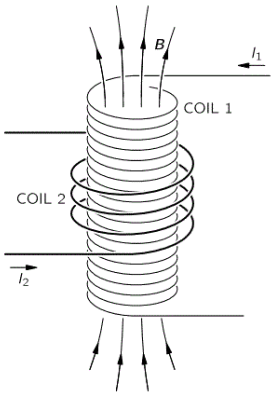
$$V_{emf}\ (\text{Secondary}) = -N_s\left[\frac{\Delta\Phi}{\Delta t}\right] = -M\left[\frac{\Delta I_p}{\Delta t}\right]$$

$$M = \frac{N_s\Phi_M}{I_p} \quad \text{Units of Henry's (H)}$$



N_s is the number of coils in the secondary.
$\Delta\Phi_s$ change in flux in secondary $\Delta\Phi = \Phi_{final} - \Phi_{intial} = A_fB_f - A_iB_i$
M is the Mutual Inductance of the system, units: henry (H)
I_p is the current in the primary coils (amps).
"-" means that the induced magnetic field opposes the external magnetic field.

Example: Coil 1 is a solenoid with N_p = 600 turns, A = 0.01 m^2 and a length of L_p = 0.2 m. Coil 1 is placed inside coil 2 with N_s = 300 turns, if the current in coil 1 goes from 0 A to 5 A in 2 ms,



A) Determine the induced voltage in coil 2 due to the changing magnetic field of coil one?

The magnetic field inside a solenoid is

$$B_P(max) = \frac{\mu_o\ IN}{L} = 0.019\ T$$

$$\Delta\Phi = \Phi_f - \Phi_i = B_fA - B_iA = 1.9 \times 10^{-4}\ \text{Wb} - 0$$

$$V_{emf}\ (\text{Secondary}) = -N_s\left[\frac{\Delta\Phi}{\Delta t}\right] = -(300)\left[\frac{1.9\ x\ 10^{-4}}{2 \times 10^{-3} s}\right] = 28.5\ V$$

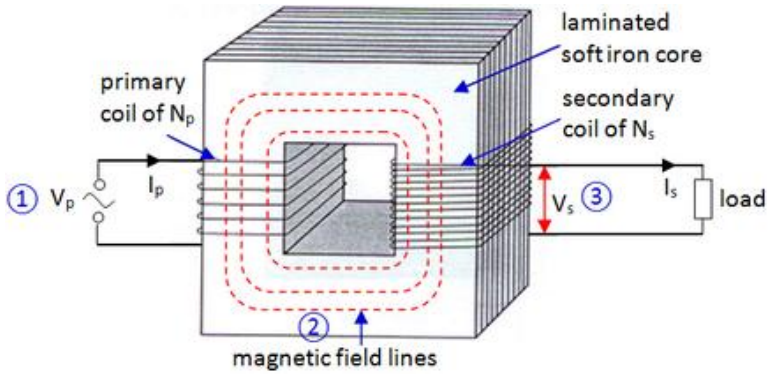
B) What is the induced voltage in coil 2 when the current in coil 1 reaches 5 A and stay at that current for 10 sec?

There is no change in magnetic flux $\Delta\Phi$ = 0 Wb therefore there is no induced voltage in coil 2.

Transformer - A device which uses mutual inductance between primary and secondary coils to increase or decrease the voltage/current from the primary to the secondary coils. Transformers transform Voltage using a changing current by different number of wire loops, turns or windings (N) on opposite sides of a ferromagnetic core material. The changing current on the primary creates the changing magnetic field in the secondary coils through the iron core which produces the induced potential.

- ✓ Transformers must have a changing magnetic field to work therefore there must be a changing current, they do not work with constant voltage/current sources.

From the diagram below:



(1) The ac power source on the primary side of the transformer.

(2) The ferromagnetic material that amplifies the magnetic field generated by the primary coil.

(3) The secondary side of the transformer, through mutual inductance of the primary and secondary coils generates a secondary voltage across the load.

Step-up transformer – the secondary voltage is larger than the primary voltage.

$N_p < N_S$ and $V_p < V_s$ for step-up transformers.

Step-down transformer – the secondary voltage is less than the primary voltage

$N_p > N_S$ and $V_p > V_s$ for step-down transformers.

The relationship between voltage V, the number of loops of wire N and the current for the primary (p) and secondary (s) are given below.

Voltage: $\frac{V_s}{V_p} = \frac{N_s}{N_p}$

Current: $\frac{I_p}{I_s} = \frac{V_s}{V_p} = \frac{N_s}{N_p}$

- A transformer that steps up the voltage simultaneously steps down the current.
- A transformer that steps down the voltage steps up the current.

- The Power in a transformer's primary and secondary are always the same, energy must be conserved!

Electric Power into primary = Electric Power out of secondary

$$I_p V_p = I_s V_s$$

Example: The primary end of a transformer is wired to an ac power source at 120 V and N_P = 200 turns and the secondary side of the transformer has N_S = 10 turns

A) What is the voltage across the secondary?

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} \; therefore \; V_s = \frac{V_p N_s}{N_p} = 6V$$

B) If the load has a resistance of 12 Ω, what current must the primary power source supply to the transformer?

$$V_s = I_s R \quad therefore \quad I_s = \frac{V_s}{R} = 0.5 \; A$$

Using the current in the secondary to determine the current in the primary

$$\frac{I_p}{I_s} = \frac{V_s}{V_p} \; \; therefore \; \; I_p = I_s \left(\frac{V_s}{V_p}\right) = 0.025 \; A$$

C) What is the electrical power supplied by the source to the primary and what is the power used by the load on the secondary?

$P_P = I_P V_P = (0.025A)(120\ V) = 3\ W$ $\quad$ $P_S = I_S V_S = (0.5A)(6\ V) = 3\ W$

Same amount of power is used by both the primary and secondary, the same Joules per second

Inductor – An electrical circuit element that is made up of coils of wire which creates the induced emf that opposes the applied potential and have many important uses in the circuit. Inductors act to limit current flow during changes in current through the circuit due to the induced emf across the coils, electric filter, sensor, voltage transformer and can be an energy storage device.

- The faster the current changes the faster the magnetic field generated by the coil will change and the larger the induced emf.

- The more coils of wire the larger the inductance due to additional $\vec{B}$-field for each coil.
- A ferromagnetic material such as iron is placed inside the coil to increase the inductance by increasing the magnetic field strength inside the coil.

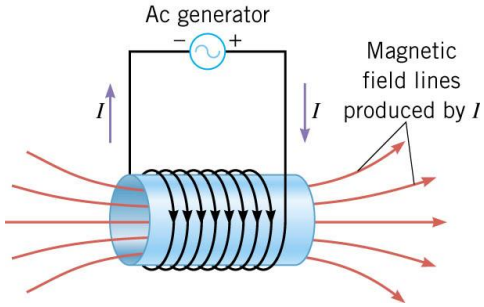


<u>Self-Inductance or Inductance</u> – The process by which a changing current in a coil of wire induces an emf in that same coil of wire to oppose the external change in current/voltage. If a changing magnetic field can induce a potential in a second coil of wire, it can also induce a potential in its own coil of wire due to the fact that the magnetic field is changing inside its coil.

<u>Emf due to self-inductance</u>

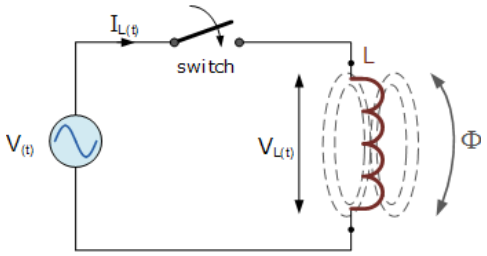
$$V_{emf} = -\frac{d\Phi_M}{dt} - L\left(\frac{\mathrm{dI}}{\mathrm{dt}}\right)$$

$$L = \frac{N\Phi}{I} \quad with\ units\ of\ Henry's(H)$$

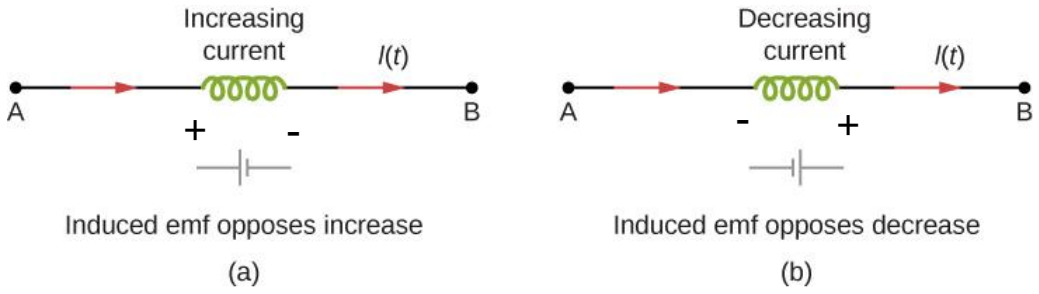


N is the number of coils, dΦ change in flux
L is the Self Inductance, units Vs/A = 1 Henry (H)
dI is the change of current in the coils
Φ is the flux through one loop

- Self-inductance is due to the changing magnetic field inside the loop which is due to the changing current in the wire.
- When there is no change in current there is no self-inductance.
- By wrapping the coil around a ferromagnetic material, the magnetic field strength and therefore the inductance can be increased substantially.
- The Inductance L is dependent on the geometry of the coil and material inside the coil.



The change in current (increasing or decreasing) in the circuit creates an induced voltage that opposes the external change. The electrical energy stored in the inductor is stored in the magnetic field. The inductor acts as a power source with a voltage that is generated that opposes the change in the applied voltage.

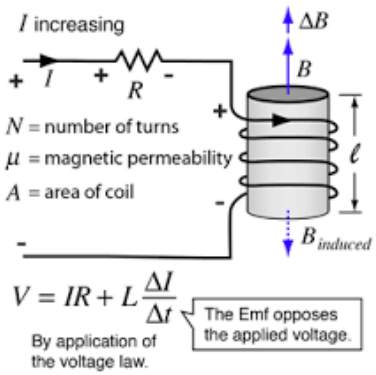


(a) The current in the circuit is moving to the right and is increasing, this induces a voltage in the inductor that opposes the increase in current which would be in the opposite direction as the external voltage.

(b) The current in the circuit is moving to the right and is decreasing, this induces a voltage in the inductor that opposes the decrease in current which would be in the same direction as the external voltage.

Inductance of a Solenoid - A solenoid is a long coil of wire with many loops therefore it experiences Self Inductance when a changing current is passed through the coils because of the changing magnetic field inside the solenoid.

$$L = \frac{\mu N^2 A}{\ell}$$



μ is the permeability of the material inside the solenoid.
N is the number of coils.
A is the cross-sectional area of the coil.
ℓ is the length of the solenoid. (lower case L).

Sometimes given as: $L = \mu n^2 A \ell$ where n = N/l is the number of coils per length

Example: Determine the self-induced voltage drop across a solenoid filed with iron (N = 50 turns, A = 5 x 10^{-4} m^2, and l = 0.1 m) wired in a circuit where the current goes from 0 A to 2 A in 10 ms? μ_{iron} = 6.5 x 10^{-3} Tm/A

$$L = \frac{\mu N^2 A}{\ell} = 0.081\ H$$

$$V_{emf}(self) = -L\left(\frac{\Delta I}{\Delta t}\right) = -(0.081\ H)\left(\frac{2\ A}{10\ x\ 10^{-3} s}\right) = 16.25\ V$$

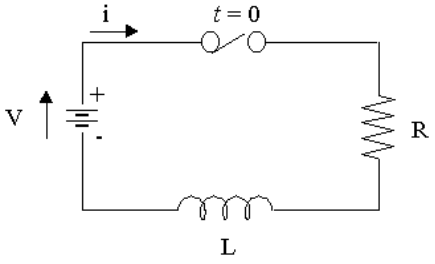
Energy stored in the magnetic field of an Inductor - When current flows through an inductor, it creates a magnetic field. The energy is stored in this magnetic field because it requires work to build up the field against the opposition caused by the inductor. An inductor resists changes in current, when the current changes, the inductor generates a voltage to oppose this change and generates the stored magnetic potential energy.

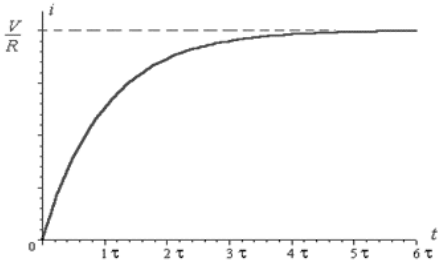
$$PE_M = \frac{1}{2}LI^2$$

The magnetic potential energy (PE_M) is proportional to the inductance (L) of the inductor and the square of the current (I) though the inductor.

RL Circuits using dc power source – An RL circuit is an electrical circuit containing a resistor (R) and an inductor (L), connected in either a series or parallel configuration, that is powered by a voltage or current dc source. These circuits resist changes in current flow through the inductor's self-inductance, which generates a magnetic field.

- When the switch is closed the current begins to flow and the inductor induces a voltage to oppose this current flow. Over time the current begins to reach a steady state and there is not as large a current change there fore not as much induced voltage to oppose it. As time the current does not change and the inductor no longer induces any voltage, the inductor acts as only a wire in the circuit.





LC circuit oscillations – An LC circuit is an electric circuit containing an inductor (L) and a charged capacitor (C), forming a resonant, tuned, or tank circuit that stores and exchanges energy between its electric field and magnetic field. This energy exchange creates oscillations at a specific resonant frequency. The ideal LC circuit conserves energy, while real-world implementations experience damping from resistance, causing the oscillations to eventually die out.

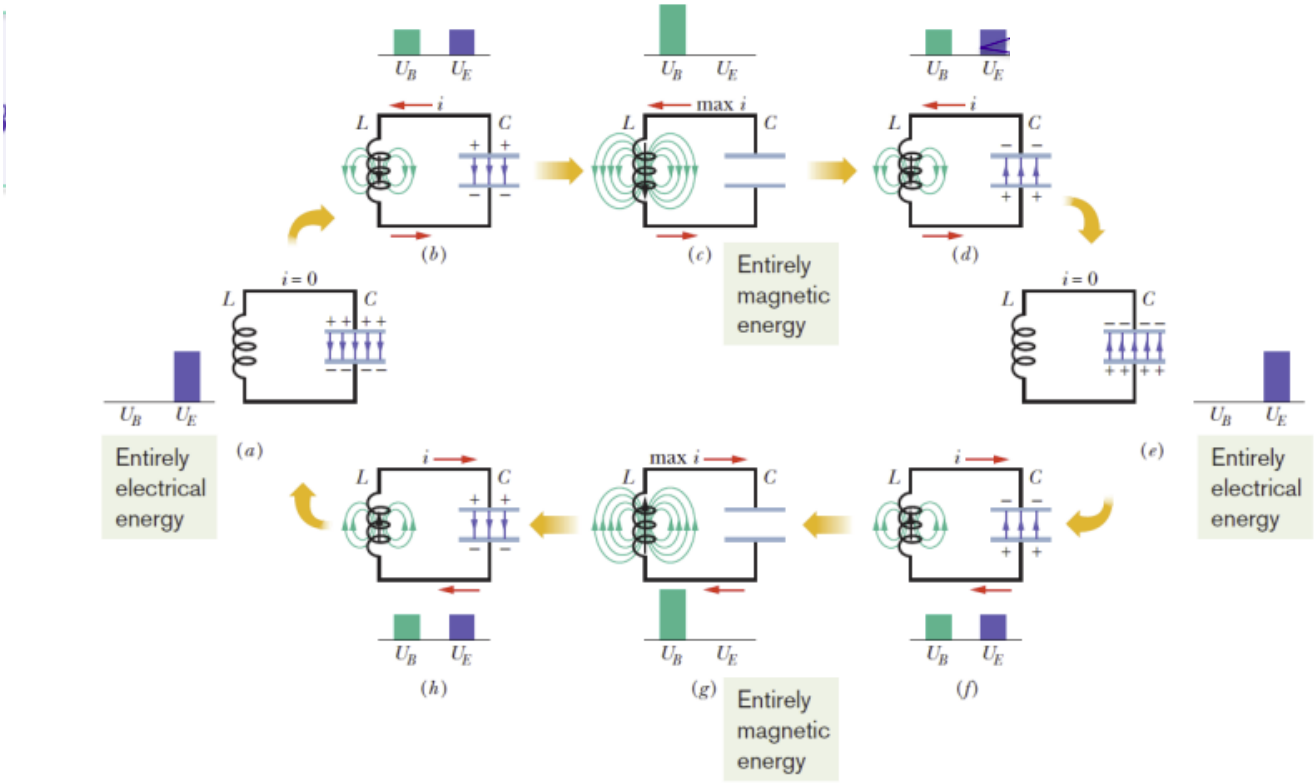
Note: In the diagram below, the magnetic potential energy $PE_M = U_B$ and the electric potential energy $PE_E = U_E$.

(a) Starting with a charged capacitor – all the energy is stored in the electric field of the capacitor and no current is flowing.

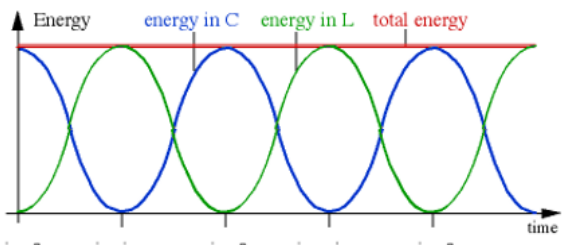
(b) As the current flows from the capacitor, the inductor senses a changing current/magnetic field and creates a voltage to oppose this change. Electrical energy is transformed from the electric field and is now being stored in the magnetic field of the inductor.

(e) The induced voltage on the inductor re-charges the capacitor from the energy in the magnetic field. Because the voltage induced on the inductor opposes the changing field, the voltage is the opposite sign as the original charged capacitor, therefore the charge on the capacitor is revised.

Now the capacitor is fully charged in the opposite direction, the cycle starts over.

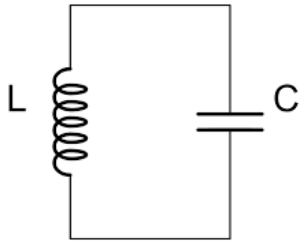


Graph of energy stored in the capacitor, in the inductor and total energy in the system for an LC circuit.



- When the voltage across the capacitor is maximum the current is zero and all the electrical energy is stored in the capacitor's electric field and none in the inductor.
- When the current through the circuit is maximum the voltage across the capacitor is zero and all the electrical energy is stored in the inductor and none in the capacitor.
- The total energy of the system does not change with time.

In an ideal LC circuit with no resistance, the energy constantly moves between the capacitor's electric field and the inductor's magnetic field, creating a sustained, simple harmonic oscillation.



The frequency of oscillation (f) for an LC circuit is called its resonant frequency. It is determined solely by the values of the inductor (in Herny's) and the capacitor (in Farads).

$$f = \frac{1}{2\pi\sqrt{LC}}$$

In a real-world circuit, some resistance is always present in the components and wires. This resistance causes the energy to dissipate over time as heat, a process known as damping. In a damped LC circuit, the amplitude of the oscillations gradually decreases until it stops.

Example: A simple LC circuit has a capacitor fully charged, the inductor has a value of 2 mH and the capacitor has a value of 6 μF.

A) Determine the frequency of oscillation of the circuit.

$$f = \frac{1}{2\pi\sqrt{LC}} = 1{,}453\ Hz$$

B) Using the same inductor of 2 mH, what capacitor must the capacitor be in order to oscillate at 1,000 Hz? Solve the resonance equation for the capacitance.

$$\sqrt{C} = \frac{1}{2\pi f\sqrt{L}} \qquad C = \frac{1}{4\pi^2 f^2 L} = 1.2\ x10^{-5} F$$

LC circuits are fundamental to electronics and telecommunications due to their ability to resonate at a specific frequency.

- Tuning circuits: In radios and televisions, an adjustable LC circuit allows you to tune in to a specific radio or TV station. The circuit is adjusted until its resonant frequency matches the transmission frequency, causing that signal to be received most strongly.
- Filters: By using the properties of series and parallel LC circuits at resonance, filters can be designed to pass or block signals of a specific frequency. A band-pass filter allows only frequencies near resonance to pass, while a band-stop filter rejects them.
- Oscillators: LC circuits are used in electronic oscillators to generate a periodic signal of a particular frequency.